



Network Analysis

AN INTRODUCTION FOR HUMANISTS

Dr Katarzyna Anna Kapitan
18 March 2026

Recap

Recap: Fundamental Network Features

New Measure: Heterogeneity

Modelling Fundamentals

Recap: Features of Real Networks

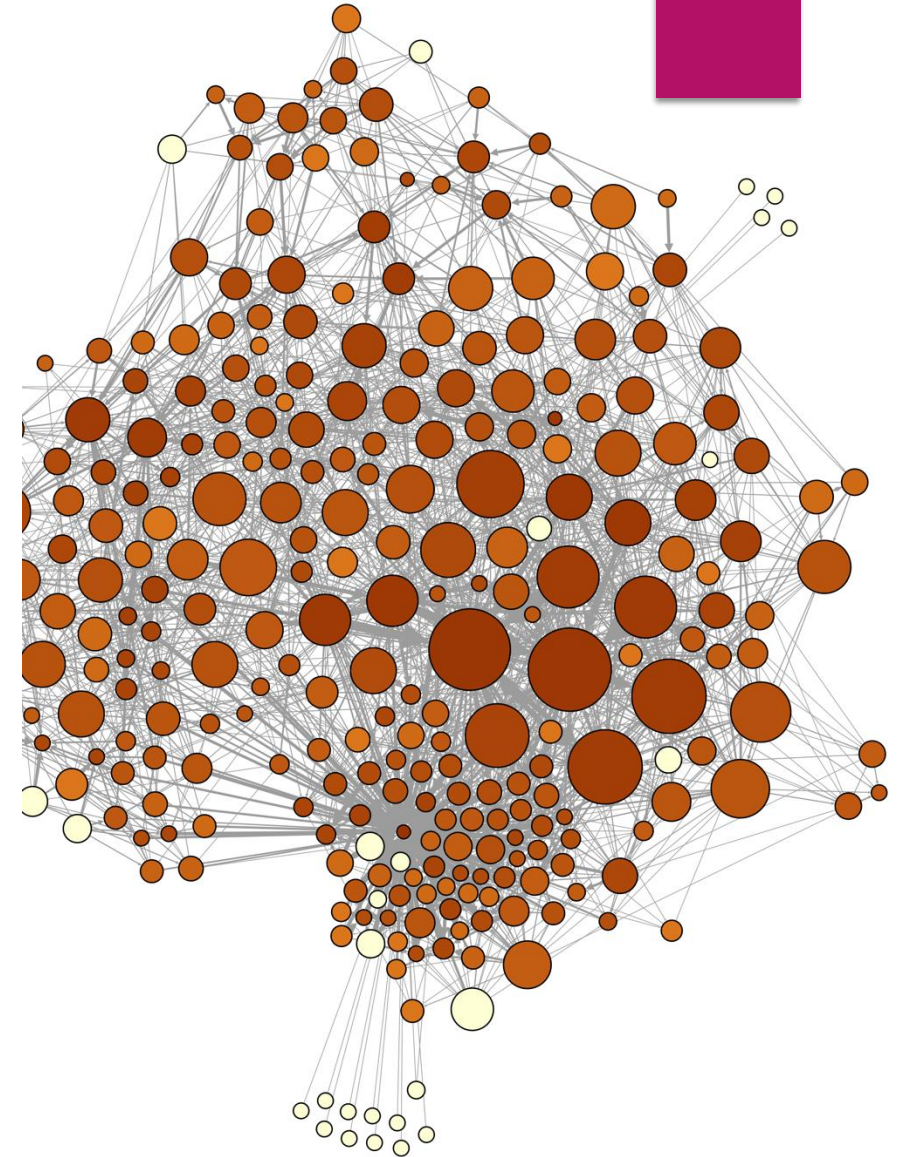
- ▶ Many real networks have:
 - ▶ Short Average Path Length
 - ▶ High Clustering Coefficient
 - ▶ Hubs



Network of Protein interaction of yeast.

Image source: Slides for Slides for Menczer, Fortunato, Davis, *A First Course in Network Science*

- ▶ Many real networks are not human-centric (!), but have instead technical, biological, systemic, or cyber-physical nature.
- ▶ Networks of protein interactions (e.g., yeast protein interactions)
 - ▶ See: Gursoy A, Keskin O, Nussinov R. 'Topological properties of protein interaction networks from a structural perspective'. *Biochem Soc Trans.* 36 (6):1398-403. doi: 10.1042/BST0361398.
- ▶ Ecological networks (e.g., everglades food-web)
 - ▶ See: Dunne JA, Williams RJ, Martinez ND, 'Food-web structure and network theory: The role of connectance and size'. *Proc. Natl. Acad. Sci. U.S.A.* 99 (20): 12917-12922. doi: 10.1073/pnas.192407699.
- ▶ Neural networks (e.g., neural network of the roundworm *c. elegans*)
 - ▶ See: Ripoll-Sánchez, Lidia et al., 'The neuropeptidergic connectome of *C. elegans*.' *Neuron.* 111 (22): 3570-3589. doi: 10.1016/j.neuron.2023.09.043.



Neural Network of *c. elegans*

Image source: Slides for Slides for Menczer, Fortunato, Davis, *A First Course in Network Science*

Short Average Path Length (Wk 3)

Network	Nodes (N)	Links (L)	Average path length ($\langle \ell \rangle$)	Clustering coefficient (C)
Facebook Northwestern Univ.	10,567	488,337	2.7	0.24
IMDB movies and stars	563,443	921,160	12.1	0
IMDB co-stars	252,999	1,015,187	6.8	0.67
Twitter US politics	18,470	48,365	5.6	0.03
Enron Email	87,273	321,918	3.6	0.12
Wikipedia math	15,220	194,103	3.9	0.31
Internet routers	190,914	607,610	7.0	0.16
US air transportation	546	2,781	3.2	0.49
World air transportation	3,179	18,617	4.0	0.49
Yeast protein interactions	1,870	2,277	6.8	0.07
C. elegans brain	297	2,345	4.0	0.29
Everglades ecological food web	69	916	2.2	0.55

Average path length of a network is the average number of steps along the shortest paths between all pairs of nodes.

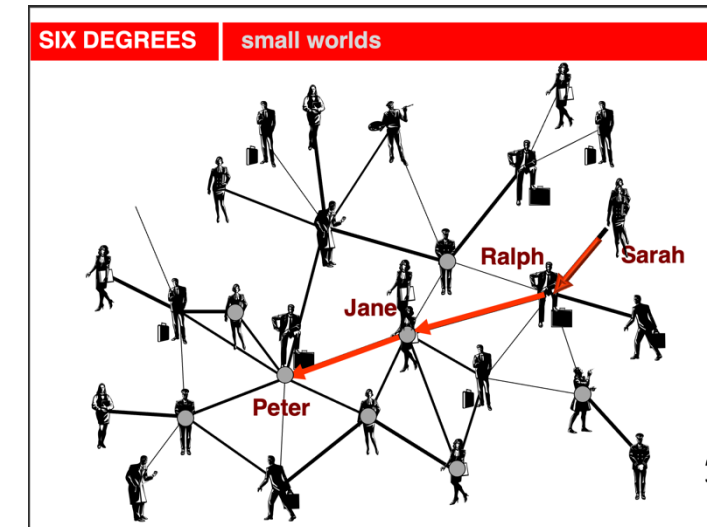


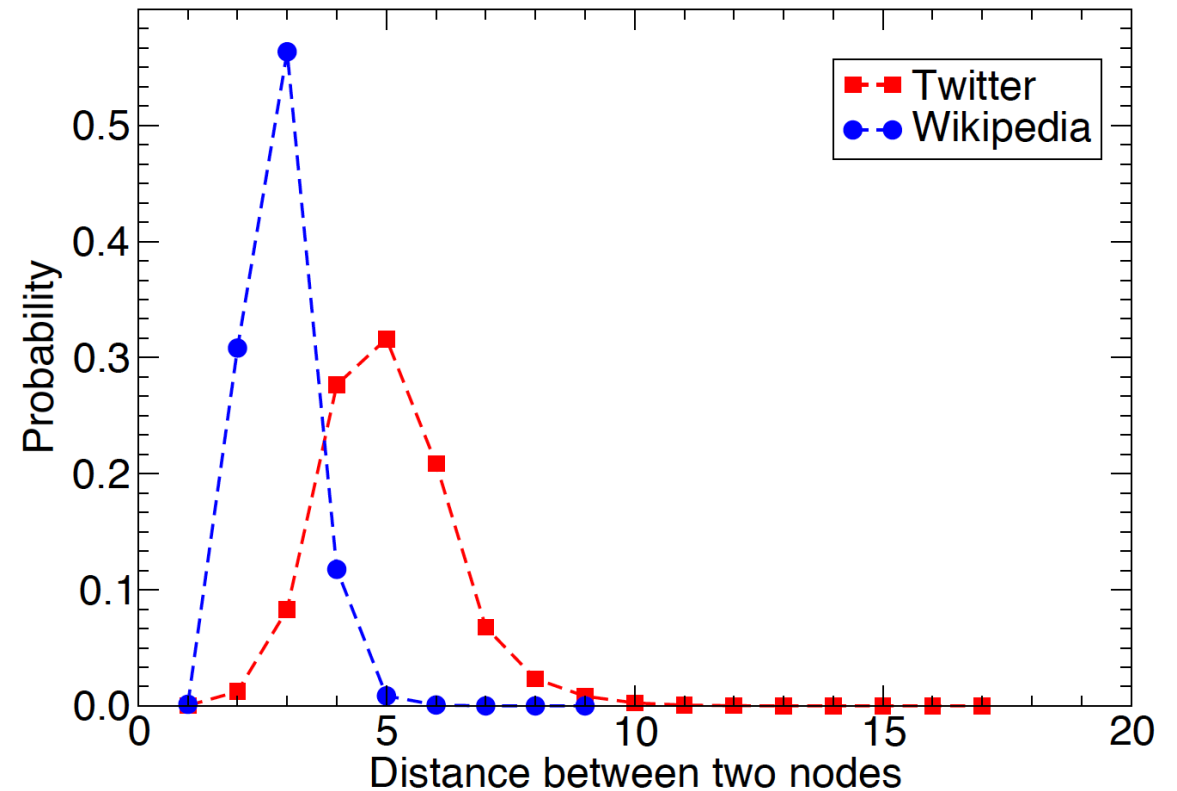
Image Source: Slides for Menczer, Fortunato, Davis, *A First Course in Network Science* (left) & Slides for Barabási, *Network Science* <https://networksciencebook.com> (right)

Short Average Path Length (Distribution)

The small-world property is typical of most real networks of interest.

If the network has hubs, paths are ultra-short.

Many shortest paths go through hubs (e.g., air transportation network)



Shortest-path length distribution

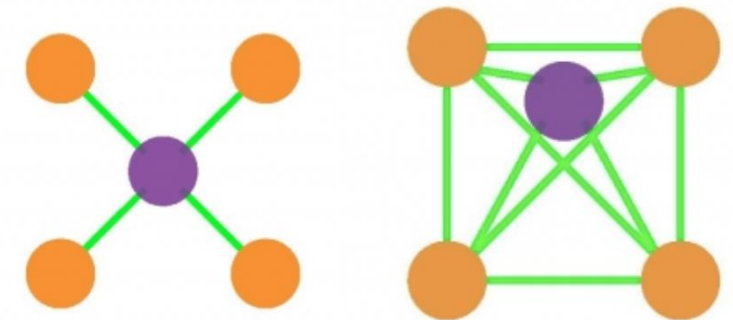
(Image Source: Slides for Menczer, Fortunato, Davis, *A First Course in Network Science*)

High Clustering Coefficient (Wk3)

Network	Nodes (N)	Links (L)	Average path length ($\langle \ell \rangle$)	Clustering coefficient (C)
Facebook Northwestern Univ.	10,567	488,337	2.7	0.24
IMDB movies and stars	563,443	921,160	12.1	0
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Image Source: Slides for Menczer, Fortunato, Davis, *A First Course in Network Science*

The clustering coefficient of a node is the fraction of pairs of the node's neighbours that are connected to each other



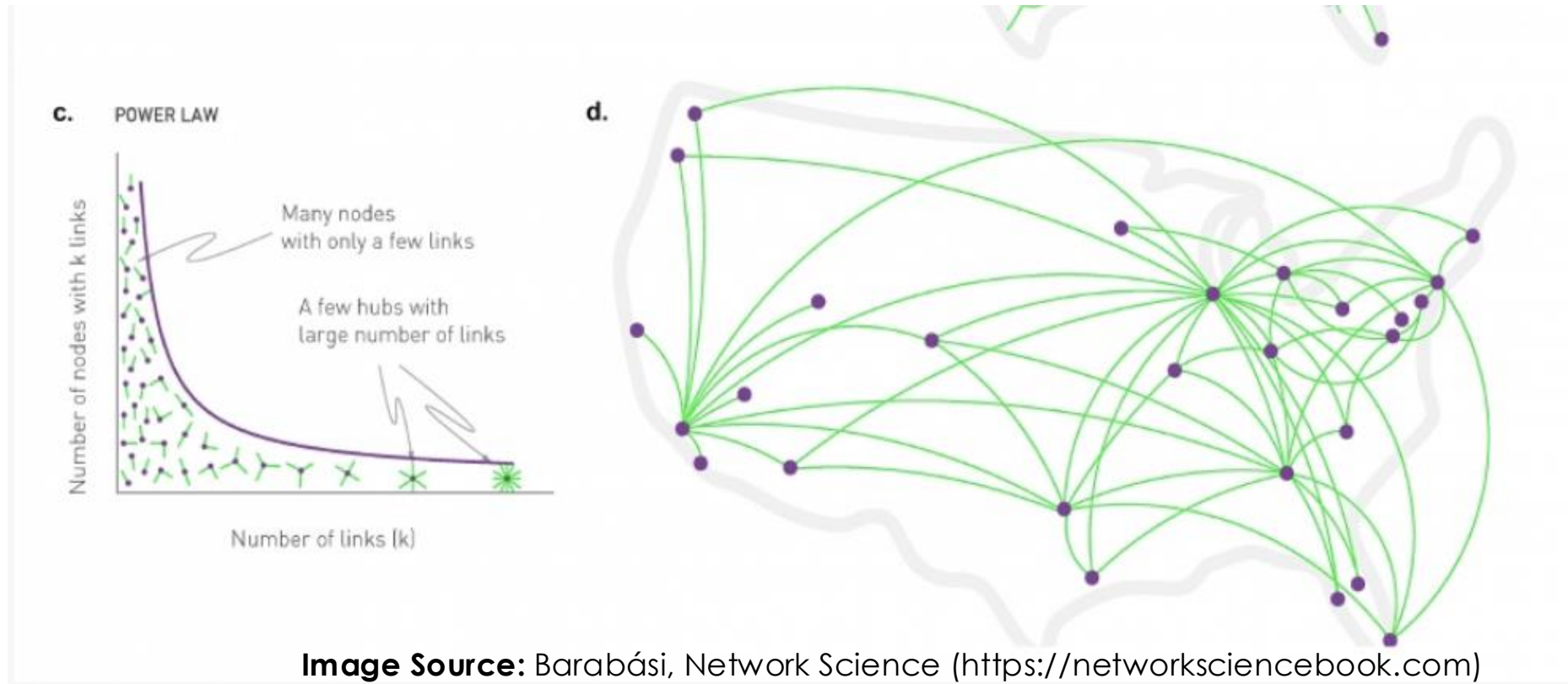
$C_i = 0$

$C_i = 1$

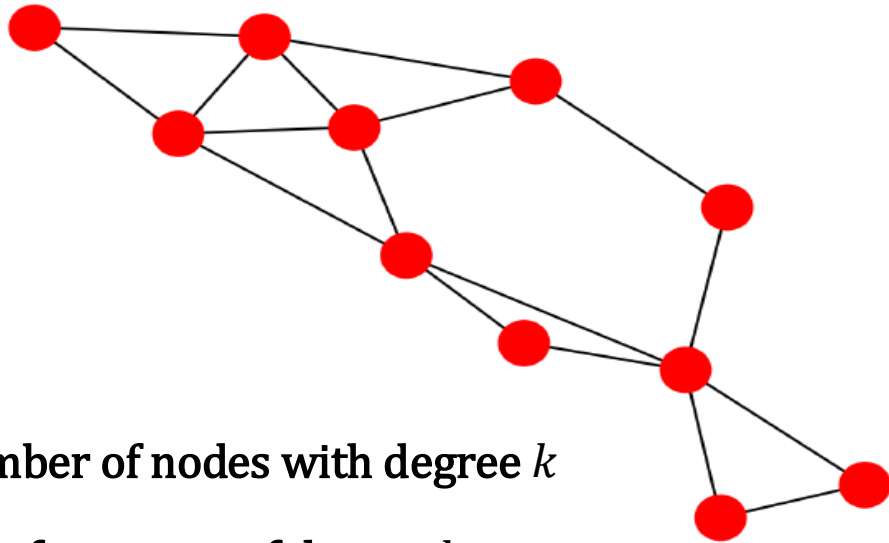
Image Source: Barabási, *Network Science*
(<https://networksciencebook.com>)

Hubs

Many real-life networks have heavy-tailed degree distribution, caused by the presence of HUBS



Hubs & Heterogeneity (Degree distribution)



n_k = number of nodes with degree k

$f_k = \frac{n_k}{N}$ = frequency of degree k

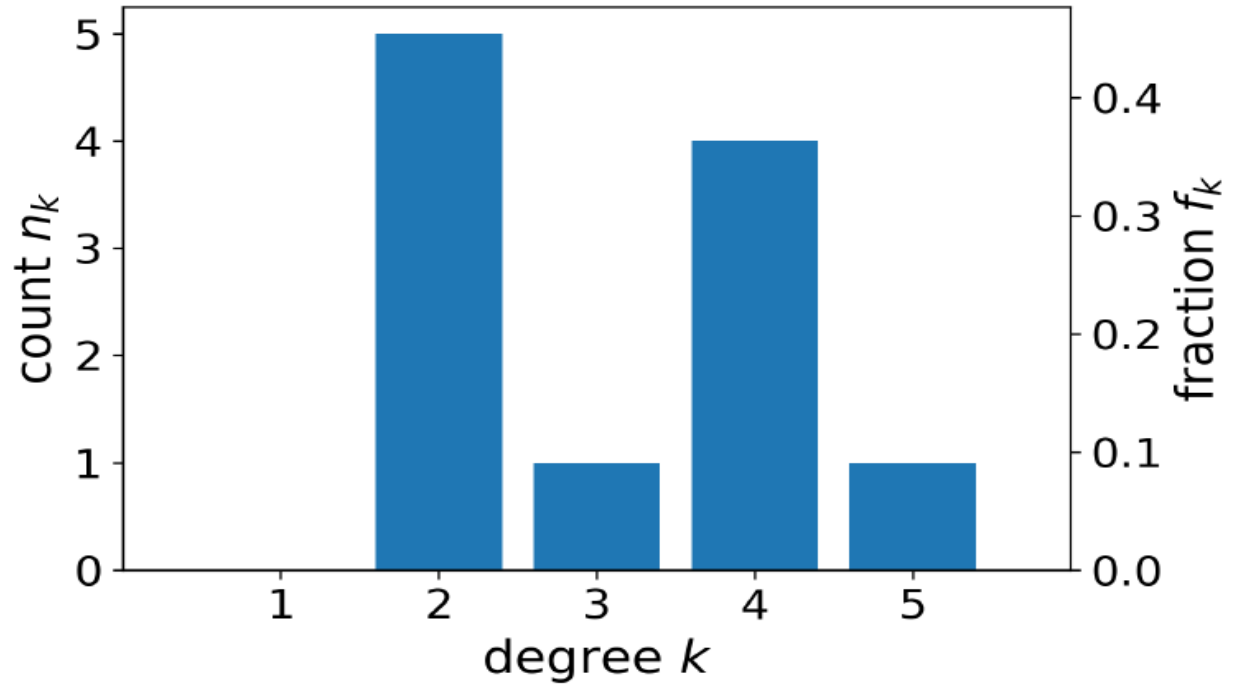


Image source: Menczer, Fortunato, Davis, *A First Course in Network Science*, version 3 (Cambridge University Press 2023)

Heterogeneity Parameter

Heterogeneity parameter κ is a measure of how broad the degree distribution is

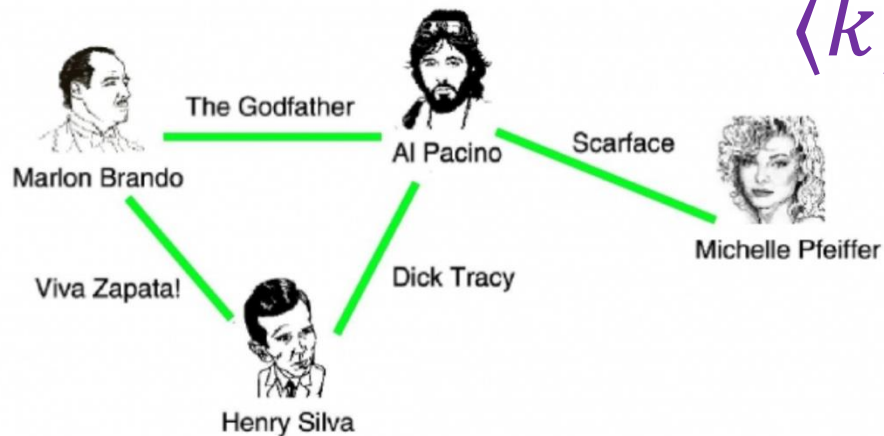
If $\kappa \approx 1$ (homogeneous)
if $\kappa \gg 1$ (heterogeneous)

Network	Nodes (N)	Links (L)	Average degree ($\langle k \rangle$)	Maximum degree (k_{max})	Heterogeneity parameter (κ)
Facebook Northwestern Univ.	10,567	488,337	92.4	2,105	1.8
IMDB movies and stars	563,443	921,160	3.3	800	5.4
IMDB co-stars	252,999	1,015,187	8.0	456	4.6
Twitter US politics	18,470	48,365	2.6	204	8.3
Enron Email	36,692	367,662	10.0	1,383	14.0
Wikipedia math	15,220	194,103	12.8	5,171	38.2
Internet routers	190,914	607,610	6.4	1,071	6.0
US air transportation	546	2,781	10.2	153	5.3
World air transportation	3,179	18,617	11.7	246	5.5
Yeast protein interactions	1,870	2,277	2.4	56	2.7
C. elegans brain	297	2,345	7.9	134	2.7
Everglades ecological food web	69	916	13.3	63	2.2

Image Source: Slides for Menczer, Fortunato, Davis, *A First Course in Network Science*

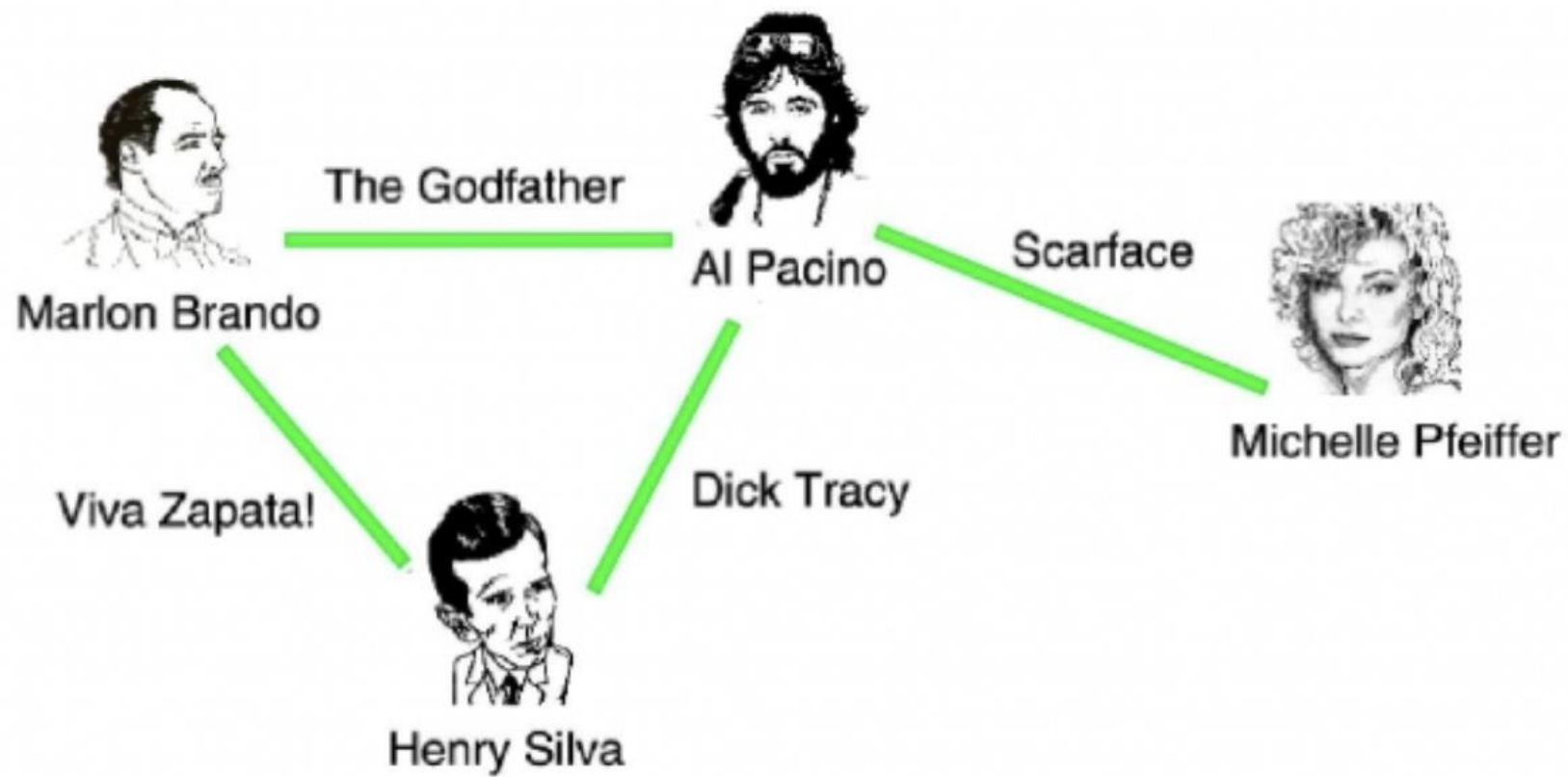
Heterogeneity Parameter

$$\kappa = \frac{\langle k^2 \rangle}{\langle k \rangle^2}$$



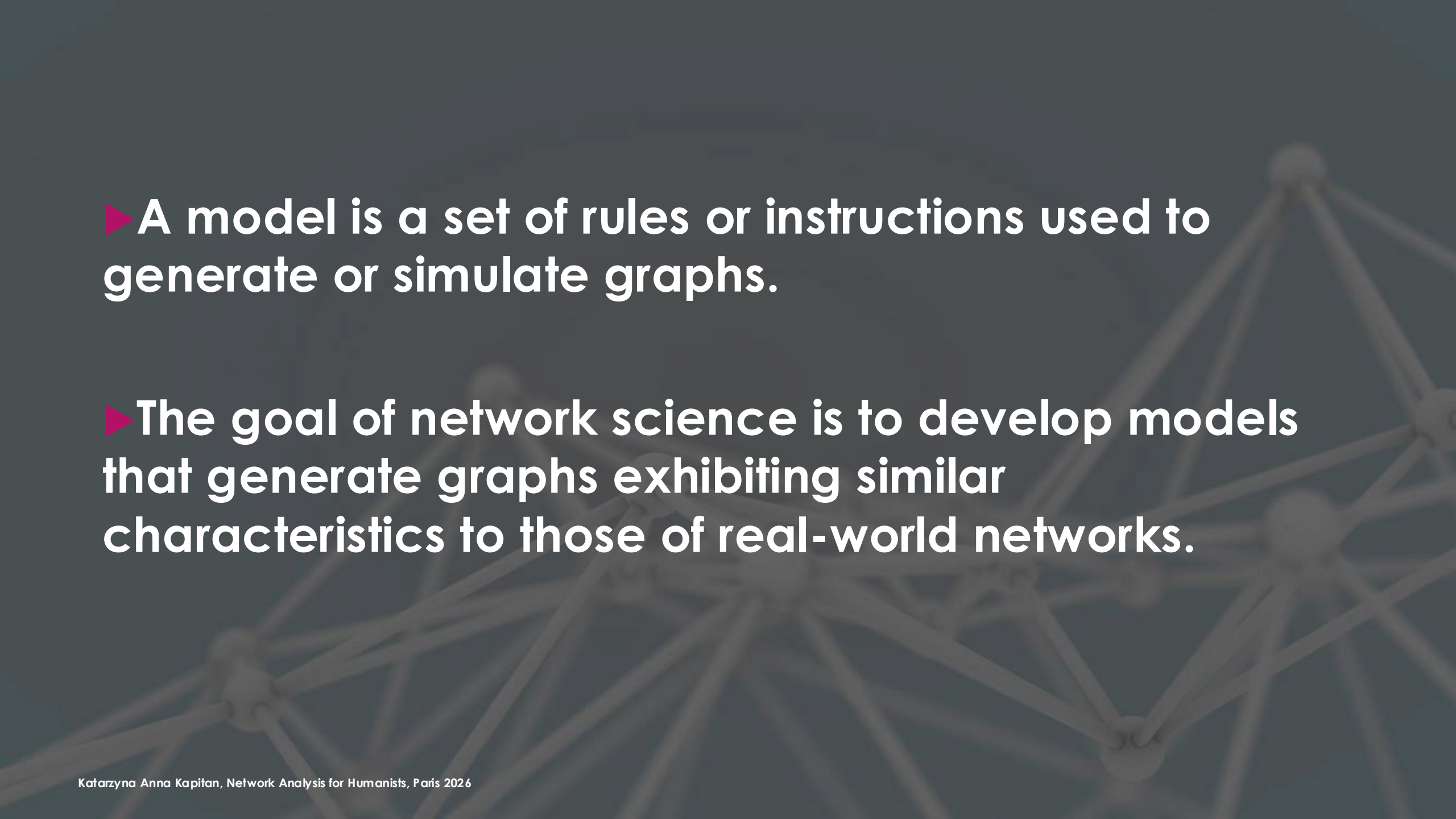
Is this actor network
heterogenous or homogeneous?
What is κ ?

- ▶ $\langle k \rangle$ - first moment of the degree distribution (average degree)
- ▶ $\langle k^2 \rangle$ - second moment of the degree distribution k^2 (average of squared degrees)
- ▶ κ - ratio of second moment to square of the average (**Heterogeneity parameter**)
- ▶ When the second moment is much larger than the square of the first moment, the network has very large hubs
=> **high heterogeneity (unequality).**



Modelling

WHAT AND WHY OF MODELLING



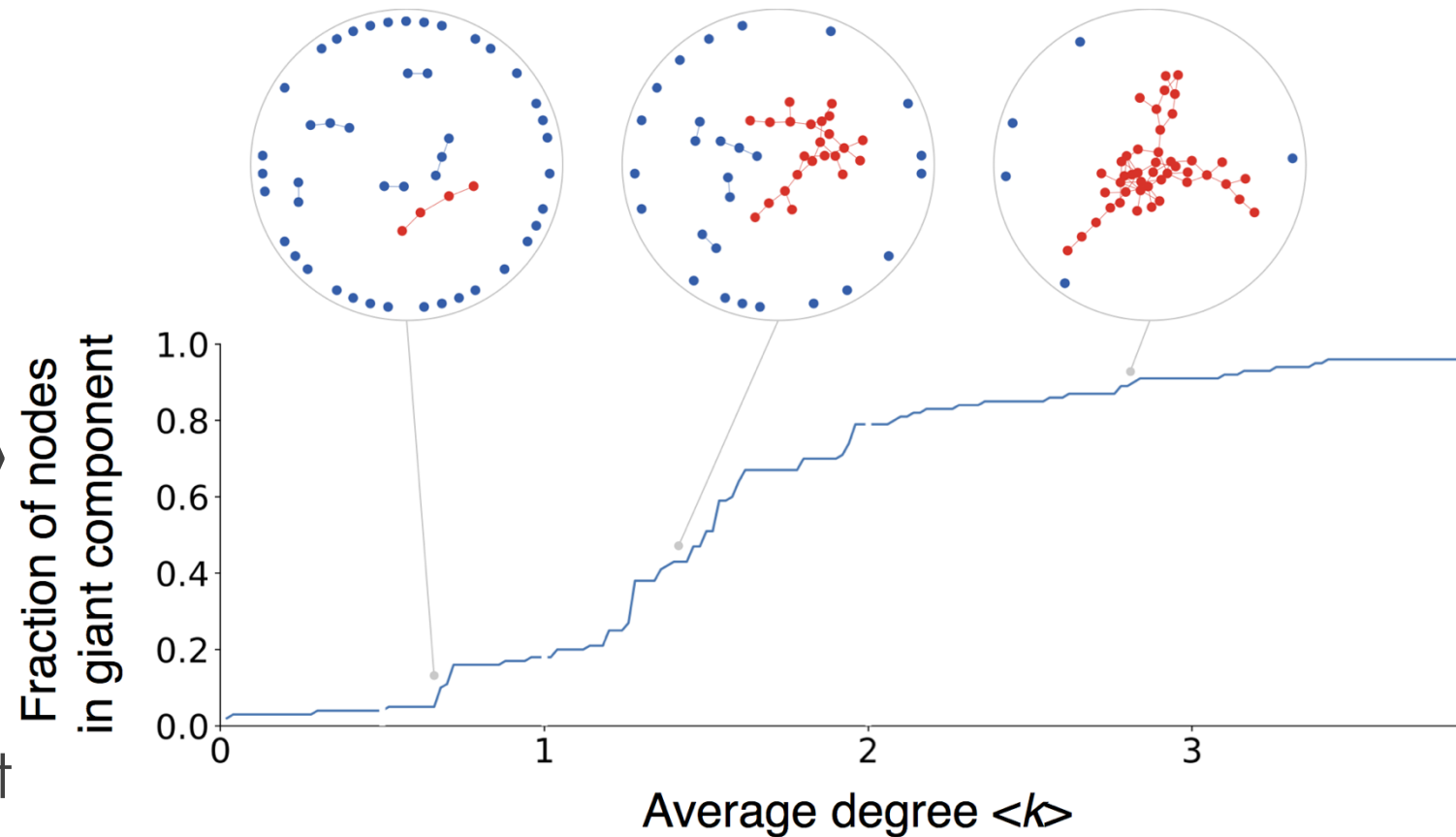
► A model is a set of rules or instructions used to generate or simulate graphs.

► The goal of network science is to develop models that generate graphs exhibiting similar characteristics to those of real-world networks.

Random Graphs (Erdős-Rényi & Gilbert models)

- ← Generate random graph by placing links at random between pairs of nodes
- ← **Algorithm:**
 - ← Start with N nodes and zero links
 - ← Go over all pairs of nodes;
 - ← for each pair of nodes i and j , generate a random number r between 0 and 1
 - If $r < p \Rightarrow i$ and j get connected
 - If $r > p \Rightarrow i$ and j remain disconnected

- ▶ In random graphs the average shortest-path length (L) depends on the number of nodes (N) and the probability of edge formation (p), which determines the average degree of the network.
- ▶ In **random graphs**, a **giant component** begins to form when the average degree $\langle k \rangle$ crosses a critical threshold determined by N .
- ▶ Around $\langle k \rangle = 1$ a giant component grows very fast at the expense of other smaller components.



Random Graphs (problems)

For a random graph to be a good model of real networks, the link probability p should be close to zero, as **real networks are usually sparse**, i.e. have small density.

If p is set close to zero then the **clustering co-efficient** is also close to zero, as in random graphs, the probability that a pair of neighbours of a node is connected is p . The link probability is the same for every pair of nodes, regardless of their having common neighbours or not.

The Watts-Strogatz model

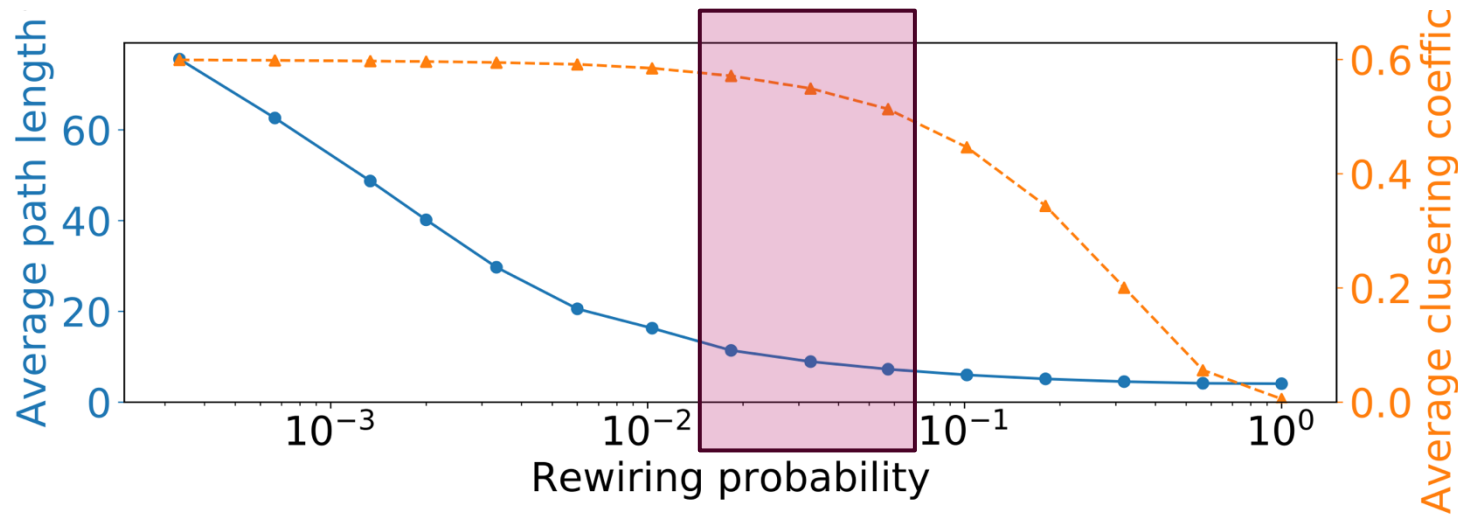
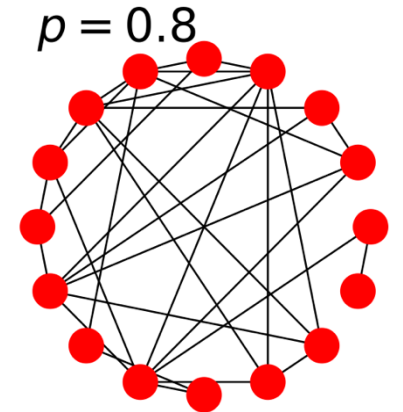
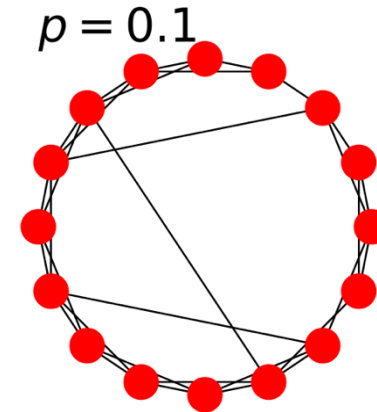


Image Source: Slides for Menczer, Fortunato, Davis, *A First Course in Network Science*

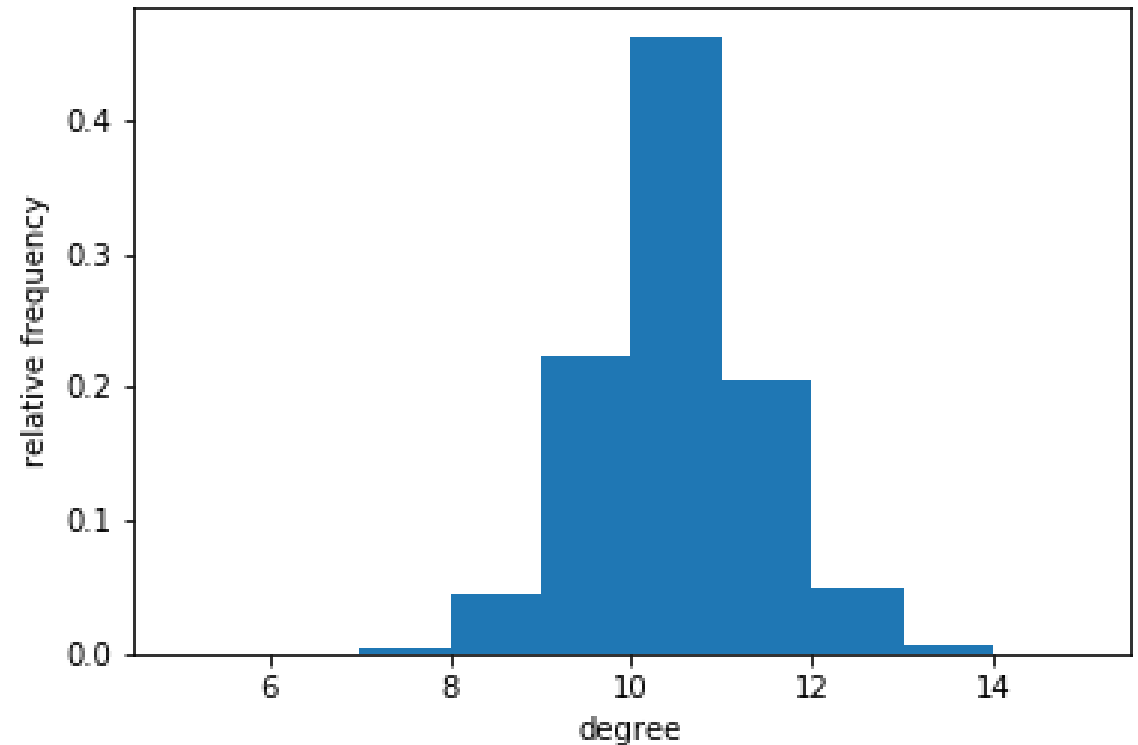


N nodes form a regular ring lattice, with even degree k .

With probability p , each link is rewired randomly

The Watts-Strogatz model (problems)

- The degree distribution is peaked as most nodes have the same degree: **no hubs!**
- The Watts-Strogatz model fails to reproduce the **broad degree distributions** observed in many real-world networks



The Barabási-Albert Model

Algorithm:

- Start with a group of m_0 nodes, usually fully connected (clique)
- At each step a new node i is added to the system, and sets m links with some of the older nodes ($m \leq m_0$) **(Growth)**
- Each new link is wired to an old node j with probability which is proportional to the degree k_j of j . **(Preferential Attachment)**

The Barabási-Albert Model

Generates graphs where a small number of nodes accumulate the majority of the links (hubs) => skewed degree distribution that fits the power-law observed in many real-world networks.

Perfect model?

What's the possible problem here?

